3

India's Climate

You learnt how to find the average daily and monthly temperature and rainfall in class VIII. Based on what you learnt, fill in the information about the weather in your area for the entire year in the following table. If needed, you can ask your teacher to help you.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Temperature (in C)												
Rainfall (in mm)												

In which months do we receive the highest rainfall and why?

What changes do we see in our surroundings when it rains?

The diversity of the weather makes our country distinctive. The main seasons in India are winter, summer and monsoon. Of these, the monsoon season is very important for us because our economy is agriculture-based and agriculture depends on the rains.

Many people die every year in the Gangetic plains because of the extreme heat and 'loo' (hot, dry wind) during summer and the cold wave during winter. Do such deaths occur in other parts of India (south, north-east and western regions)? The rainfall in a day in the north-eastern state of Meghalaya is equal to the rainfall in 10 years in Jaisalmer district of Rajasthan. The temperature drops to -45^{0}C (45 degrees below 0) at night in December in the town of Dras in Jammu and Kashmir. During this month in Chennai in Tamil Nadu, the night temperature is 22-25^{0}C.

Have you noticed that the direction of the wind in the area where you live changes in different months of the year? Sometimes, the wind blows from the north or north-east and sometimes from the south or south-west. When there are floods in one part of India, there is drought in another part. Why does this happen? To understand why this happens, we need to first understand the processes that happen in our atmosphere.

Weather and Climate

The changes that occur in atmospheric conditions (temperature, atmospheric pressure, wind, humidity, clouds, and rainfall) in any place in a short span of a few minutes, days or month is known as the weather. The weather changes constantly. It can change many times in a day, so the morning can be very different from the afternoon.

Climate is the weather conditions over a long duration of time. Long duration means the average weather conditions over 'more than 30 years'. We can understand the climate of a place by looking at the weather statistics of that place. For example, if we add the maximum and minimum temperature during a day in that place and divide it by two, we get the average day temperature of that place. In the same way, we can calculate the average monthly temperature by adding the daily average temperature during the month and dividing the total by 30 or 31 (the number of days in that month). If we now add the average temperature of each month of the year and divide the total by 12, we get the average annual temperature. If we then add the average annual temperature of 31 years and divide the total by 31, we get the average temperature for 31 years.

Temperature is one component of climate. Other components include air pressure, wind, humidity, cloud cover, and rainfall. We can find out the climate of a place by studying all these components together. Temperature and rainfall directly impact agriculture, industry, transportation, construction, etc.

Write the formula to find the average daily, monthly and annual temperature.

Factors that Control the Climate

The earth's surface is very vast and varied. There are continents and oceans. The surface of the continents is not the same everywhere. Some places have high mountains, some have plains. Some places have dense forests, some have vast deserts. The oceans are also not evenly spread across the earth's surface. Water covers 39% of the northern hemisphere and 81% of the southern hemisphere. These surface differences influence the process of heating and cooling of the atmosphere. This uneven heating and cooling of the atmosphere is what controls the climate. The main causes of climatic variation in the Indian subcontinent are the following:

1. Latitudinal position
2. Altitude
3. Distance from the sea
4. Air pressure
5. The Himalayan range
6. Human factors

1. Latitudinal Position

The sun's rays fall vertically on the Equator throughout the year. As we move from the Equator to the Poles, the sun's rays spread over a larger area of the earth's surface because the earth is spherical (see Figure 3.1). So the heat energy received per unit area becomes lesser as we move away from the Equator. As a result, the temperature is lower. Thus, the distribution of the sun's heat depends on the latitude. The temperature is highest near the Equator and lowest near the Poles.

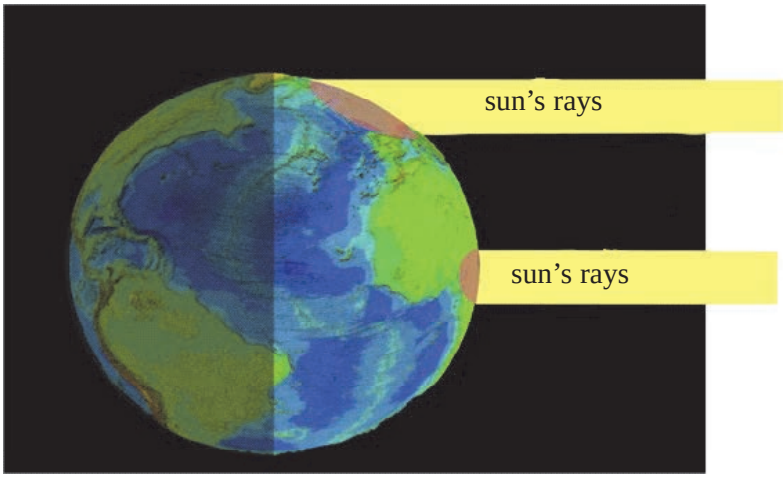


Figure 3.1: The spread of the sun's rays on the earth's surface according to the latitude

The Tropic of Cancer passes through the middle of India and divides the country into two parts. The southern part (South India) is closest to the Equator while the northern part (North India) is further

away from the Equator. This is why South India experiences tropical heat throughout the year while North India is very hot during summer and very cold during winter.

2. Altitude

As we move higher from the earth's surface, the temperature falls. The atmosphere is heated by the earth's radiation (the heat of the sun's rays reflected from the earth's surface). The lower part of the atmosphere closer to the earth's surface is heated more by this radiation than the upper part. The water vapour, dust particles, and gases present in the lower layer of the atmosphere absorb the earth's radiated heat. So the temperature near the surface of the earth is higher. Because the upper layer does not have these particles, its temperature is much lower. The temperature falls by one degree centigrade on average for every 165 metres of height.

As we climb to higher altitudes, the air becomes more rarefied (less dense). The air near the earth's surface is heated more by the earth's radiation than by the direct rays of the sun. That is why people visit places like Leh, Shimla, Mussoorie and Mainpat in summer because the temperature is lower in the hills.

Radiation is the heat waves that are emitted by a hot object. The earth gets the most heat from the sun at 12 noon but we feel the maximum heat at around 2pm because the heat waves radiate from the surface of the earth at that time.

3. Distance from the Sea

The temperature does not change much in coastal areas throughout the year, so the climate is even, or less variable. But in areas that are far from the sea the climate is more variable. We learnt in class VIII that an even (less variable) climate means there is very little difference between the average temperature in summer and winter. In a variable climate, however, there is a significant difference between the average temperature in summer and winter. Why does this happen? Water takes a long time to heat and also cools down very slowly. But land heats up quickly and also cools quickly. So in coastal regions, the sea breeze keeps the temperature constant both in winter and summer. But the sea breeze has no effect on land far from the coast. So the temperature difference between summer and winter is higher in these places.

4. The Himalayan Mountain Range

The Himalayas and other mountain ranges spread across the northern, north-western and north-eastern parts of India. These high mountain ranges form a barrier that blocks the cool winds blowing towards India from across Central Asia. They also prevent the monsoon winds from the south-west from blowing out of India. In this way, the location of the Himalayan mountain range in the Indian subcontinent gives India a special kind of climate which is known as the 'tropical monsoon climate'.

5. Air Pressure

Air pressure is related to temperature. When the temperature is high the air pressure is low and when the temperature is low, the air pressure is high. Winds always blow from high pressure to low pressure areas. Why does this happen? Low air pressure means the air is less dense (rarefied). The air over any land area rises when it is heated. So the area has less air; it is less dense. Suppose a cubic meter of air in one area contains 1,00,000 molecules and a cubic meter of air in another place contains

1,10,000 molecules. We will then say the air pressure in the first area is lower than the air pressure in the second area.

Human Factors

Human economic activities, industrialization, urbanization, changes in land use, deforestation, etc cause the atmospheric temperature to rise. This impacts the climate and leads to climate change.

The average temperature of different places (^{0}C)

Place	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
Mumbai	25	24	26	28	30	29	27	27	27	28	27	25
Jagdalpur	20	23	27	29	31	29	25	25	26	25	21	19
Shimla	5	5	10	15	18	20	18	18	16	10	10	11
Delhi	14	17	23	29	34	34	31	30	29	26	20	16

Answer the following questions on the basis of the data contained in the table:

1 Which are the coldest and hottest months in Delhi?

2. Which of the above mentioned places have even or less variable climate?

3. Calculate the average annual temperature of Jagdalpur.

4. Which of these cities has the lowest average temperature?

5. Compare the temperature of Mumbai and Jagdalpur.

Seasons

The weather department of India has divided the year into four seasons:

1. Summer 2. Rains (Monsoon) 3. Autumn 4. Winter

1. Summer

The summer falls between March and May in India. The sun shines directly on the Equator on March 21st. The temperature keeps rising in the northern hemisphere from March until May. On June 21st, the sun is directly over the Tropic of Cancer. The temperature is at its peak in North India at this time. Because of the high temperature, a low pressure belt is created from the Thar Desert in the west to the Gangetic plains in the east. From March to May, the hot and dry westerly winds blow across North India during the day. They slow down at night and blow in different directions. These winds are called '*loo*'. These winds blow across Punjab, Haryana, Uttar Pradesh, Bihar and other parts of the northern plains. During these months, these areas experience fierce dust storms accompanied by showers that help to lower the temperature. These dust storms have different names in different places. They are called *'kalboishakhi'* in West Bengal and they are called *'amravrishti'* in South India. Sometimes, the hot winds encounter the cool breeze from the sea to cause cyclones accompanied by strong winds and rain (see Reference map 4).

2. Rains (Monsoon)

The rainy season falls between June and September in India. About 85-90% of the rainfall across India is experienced during these months. Along with India, Sri Lanka, Bangladesh, Myanmar, Nepal and Pakistan also get rainfall during these months. At this time, the temperature, air pressure, wind direction and the rainfall pattern is different from the other seasons. The temperature falls because of the moisture in the air. Agricultural work begins at this time (see Reference map 5).

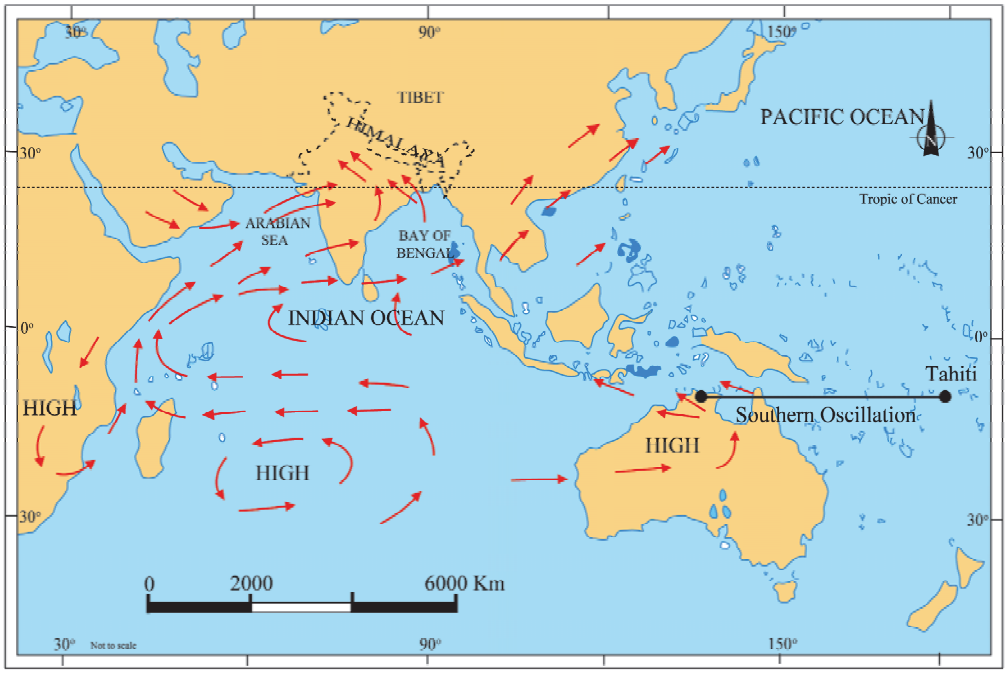
South-west Monsoon: The rainfall between June and September in India is known as the monsoon. The word monsoon was first used in India. In earlier centuries, Arab traders sailed by ship with the help of the winds that blew across the Arabian Sea. They called these winds '*mausim*', after the Arabic word for season. So the winds that brought rain to India every year came to be known as the monsoon. These winds blow from the south-west direction so the monsoon is called the 'south-west monsoon'. The monsoon spreads across India from the south-west direction. When the monsoon winds reach the north-east, they turn and blow in a north-westerly direction.

The question to be answered is: what causes the south-west monsoon? There are several causative factors which we shall now discuss:

1. Due to the excessive summer heat in May-June, a low pressure area is created in the atmosphere stretching across Peshawar (Pakistan), Afghanistan and Rajasthan. At this same time, a high pressure

Ferrel's Law

According to Ferrel's Law, if we face the direction in which the wind blows (or with our back in the direction from which the wind blows), then the wind will turn towards the right in northern hemisphere and towards the left in the southern hemisphere. This rule applies to all objects in motion.



Map 3.1: Monsoon winds



Map 3.2: ITCZ

area forms in the southern hemisphere, stretching from the island of Madagascar up to the coast of Western Australia. The Arabian Sea is also a comparatively high pressure area at this time. As a result, the south-east trade winds cannot blow northwards. After May, air pressure declines over the Arabian Sea. Once this happens, the south-east trade winds blow rapidly towards the low pressure area in the north-west. They blow across the ocean at 30 km per hour. So they are moisture-laden. As they cross the Equator from the southern hemisphere, they change direction according to Ferrel's law (see box), turning right and blowing in a south-westerly direction. These winds are now called the south-west monsoon winds. The moisture-laden winds bring heavy rainfall to India.

2. When the earth's surface heats up, the air around the Equator heats up. Due to this, a low pressure zone is created in the atmosphere near the Equator. The winds blowing from the northern and southern hemispheres converge at this low pressure tropical zone. Hence, this zone is known as Inter-Tropical Convergence Zone (ITCZ).

The sun's rays fall directly on different places between the Tropic of Cancer and Tropic of Capricorn throughout the year. The low pressure zone around the Equator also keeps changing its position. In June, the sun's rays fall directly on the Tropic of Cancer, so the low pressure zone shifts to the north from the Equator. In December, the sun's rays fall directly on the Tropic of Capricorn, so the low pressure zone shifts to the southern hemisphere. This shift attracts the trade winds from the southern hemisphere.

3. At this same time, in a higher layer of the atmosphere – at a height of 10-16km above sea level – winds begin to blow at a very high speed in a direction from east to west. These high-speed, high-altitude winds help the flow of the surface monsoon winds in the opposite direction from west to east. The monsoon does not arrive all over India at the same time. Its onset is different in different places. It first arrives in Kerala on June 1st, then Mumbai on June 7th, and reaches Chhattisgarh around June 15th. When it reaches the southern peninsula of India, the south west monsoon divides into two branches. One branch moves towards the Arabian Sea and the other branch moves towards the Bay of Bengal.

Arabian Sea Monsoon: In the branch that moves towards the Arabian Sea, the monsoon winds blow from south-west to the north-east. Kerala, Karnataka, Maharashtra, Tamil Nadu and Chhattisgarh get rainfall from this branch. The monsoon winds first hit the Western Ghats, causing heavy rain along the western coastal plain. By the time they cross the Western Ghats, they lose their moisture, becoming dry and hot as they move towards the east. This region, called a rain shadow region, receives scanty rainfall. Mumbai gets around 200cm of rainfall between June and September while Pune gets only 50cm.

The Arabian Sea monsoon now divides into two branches, one branch moving eastwards along the Narmada Valley to the Chota Nagpur Plateau, where it meets the Bay of Bengal monsoon branch, causing heavy rainfall in the region. The other northern branch goes through Kathiawar in Kutch to the deserts of Rajasthan. The Aravalli Hills run parallel to the monsoon winds so they cannot stop them, the average rainfall in this area being only 20cm. The winds thus blow unhindered through Gujarat and Rajasthan to the Himalayas. The air at this high altitude is hot and dry, so the monsoon loses its moisture. Hence, rainfall in the Himalayan region is low from this northern branch.

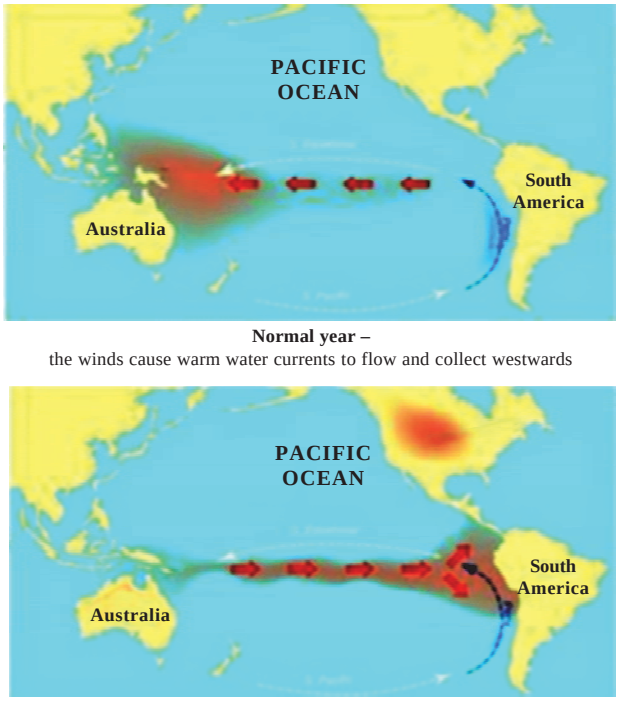
Bay of Bengal: This branch of monsoon also divides further into two branches. The first branch of monsoon winds blow to the Myanmar coast where they encounter the Arakan Yoma Mountains. The rainfall is heavy in the western range of the mountains. The other branch moves towards Bengal and Assam, with heavy rainfall in the Khasi and Garo hills. Mawsynram in Meghalaya receives the highest rainfall in the world. The winds of both branches then blow towards the west after encountering the Himalayas and the Arakan Yoma Mountains, but the rainfall is scant. Thus, Guwahati, Patna, Allahabad, Bharatpur and Jaisalmer receive scanty rainfall from the Bay of Bengal monsoon.

3. Autumn or Retreating Monsoon

In the second half of October, the temperature starts falling rapidly in north India. By the beginning of November, the low pressure zone over north-west India shifts to the Bay of Bengal. This shift gives rise to low pressure cyclonic conditions that develop into cyclones in the Andaman Sea. These cyclones generally cross the eastern coast of India. They are sometimes very destructive and cause heavy rainfall. The densely populated delta areas of Krishna, Kaveri and Godavari rivers are often hit by cyclones. They sometimes also reach the coastal areas of Odisha, West Bengal and Bangladesh. Most of the rainfall on the Coromandel Coast is caused by these cyclones. In October, the average temperature in most parts of India is 25-26^{0}C. The average temperature in Rajasthan, Gujarat and Eastern coastal plains is over 27^{0}C and in Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Madhya Pradesh, and Karnataka is below 25^{0}C (see Reference map 6).

4. Winter

The winter season extends from December to February. It is marked by a steep fall in temperature in North India. In January and February, the temperature falls below freezing point in Punjab and Kashmir. But in South India, the average temperature is 14-15^{0}C. Thus, wheat is grown in Punjab, Haryana and Uttar Pradesh during this season while rice is cultivated in Kerala and Tamil Nadu. Cyclonic storms bring heavy rainfall to Tamil Nadu and the Andaman and Nicobar Islands during this period. In winter, Kashmir and the north-western parts of India experience winter rainfall and snowfall caused by the Mediterranean Sea cyclones. North India is covered in dense fog during these months (see Reference map 3 for winter temperatures).



Normal year –
the winds cause warm water currents to flow and collect westwards

El Nino year –
warm water currents flow towards Peru due to weak eastern winds

Map 3.3: El Nino and the monsoon

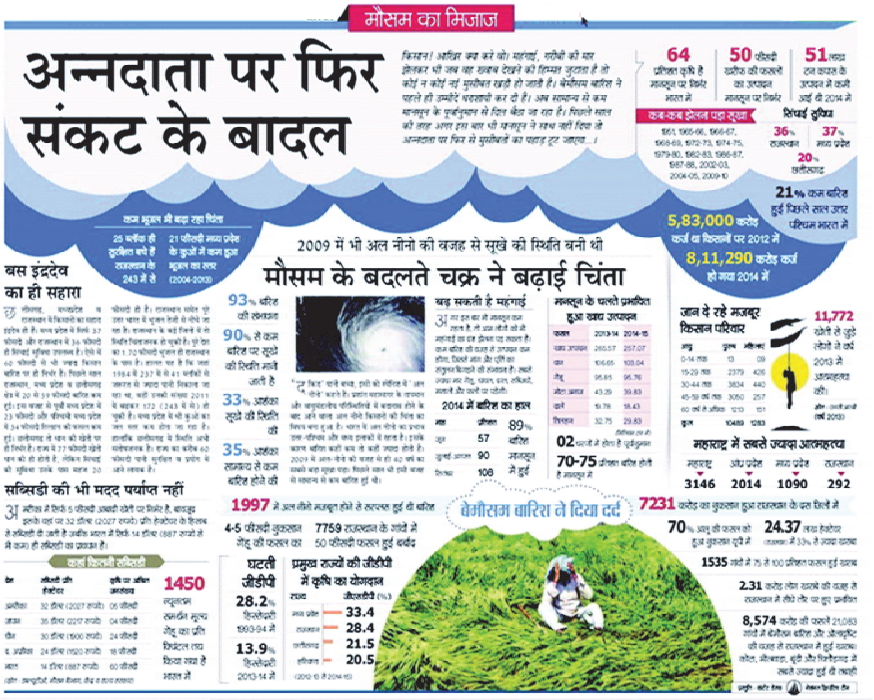
El Nino and the Monsoon: The Indian monsoon is affected by the ocean currents thousands of kilometers away near the western coast of Ecuador in South America. The heating and cooling of the Pacific Ocean as a result of these currents causes atmospheric changes that affect the monsoon.

These atmospheric changes cause the tropical convergence zone to shift 10-15 degrees southwards. The effect of this shift can be seen in the Indian Ocean. The trade winds of the southern hemisphere slow down perceptibly when they cross the low pressure convergence zone at the Equator and move north. Since these winds form the south-west monsoon, the Indian monsoon weakens.

The abnormal warming and cooling of the Pacific Ocean caused by the ocean currents occur after intervals of a few years. Peru gets heavy rainfall in winter as a result and even experiences floods. When the ocean water is warmer than normal, the fish catch decreases. This phenomenon is called El Nino (Baby Jesus) because it occurs around Christmas. When the opposite happens – the eastern Pacific Ocean cools down – it is called La Nina (the girl). (See Map 3.3 to understand this phenomenon.) El Nino is a key factor in making long term weather forecasts. In 1990-91, there was a delay of 5-12 days in the arrival of the monsoon in India due to El Nino.

Climate Change

The entire world is worried about the long term impact of the changes occurring in the composition of the earth's atmosphere. We learnt earlier that long-term atmospheric conditions is what we call climate. This means climate is a cycle of atmospheric conditions over many years. This cycle is changing. It is these changes that are causing concern. The changes


मौसम का मिजाज

अन्नदाता पर फिर संकट के बादल

मौसम के बदलते चक्र ने बढ़ाई चिंता

बस इंद्रदेव का ही सहारा

सब्सिडी की भी मदद पर्याप्त नहीं

बेमौसम बारिश ने दिया दर्द

Figure 3.2: News related to climate change in a newspaper

occur very slowly but they can lead to many problems. These problems include unpredictability of rainfall, shrinking of glaciers and rising sea levels. We need to seriously think about how a change in the date of arrival of the rainy season could affect crop cycles and land use in agriculture.

Collect news related to the weather from newspapers.

EXERCISES

1. Choose the correct option:

   i. Which of the following does not impact the climate?

   a. Latitude b. Climatologist

   c. Distance from the sea d. Air pressure

   ii In which month does Chhattisgarh experience the highest temperature?

   a. January b. March

   c. May d. November

   iii When does the monsoon arrive in India?

   a. June b. August

   c. December d. None of the above

   iv Usually, temperature …… with height.

   a. Increases b. Decreases

   c. Doesn't change d. None of the above

   v. Which country has the place which receives the highest rainfall in the world?

   a. Brazil b. Indonesia

   c. Kenya d. India

2. Explain what is climate. How is climate different from weather?
3. Name the elements associated with weather and climate.
4. What are the reasons for the changes we see in the weather where we live? Write about the changes you have experienced.
5. Which countries get rainfall during the south-west monsoon? Find out from the map and from your teacher.
6. Temperature and rainfall are the important elements of climate. How do they affect us?
7. How many main seasons do you think there are? Describe the natural and human activities that take place during these seasons.
8. Explain the arrival and departure of the monsoon.
9. Suppose the earth was flat and there were no mountains, what would happen?
10. What would happen if:

    a. It is summer all year round. b. It rains all year round.

    c. It is autumn all year round. d. It is winter all year round.

11. The average monthly temperature (in ^{0}C) and rainfall (in mm) of Bilaspur is given below. Draw a graph of these figures.

Name of the month	Average temperature (^{0}C)	Rainfall (mm)
January	17	20
February	19	30
March	23	20
April	28	20
May	35	20
June	32	200
July	25	370
August	25	380
September	25	200
October	23	70
November	19	10
December	17	0

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